WORKED SOLUTION

STEP I 1992, Question 8

Pure • Fundamental Theorem of Calculus

STEP I 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q8

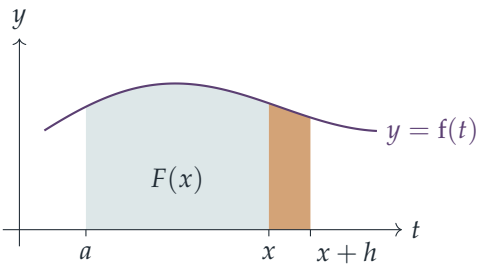
The key idea

An integral with a variable upper limit is an area that grows as the limit moves, and it grows at a rate equal to the height of the curve at that limit. That one fact (the Fundamental Theorem of Calculus) turns each *integral* equation in the question into a *differential* equation. Differentiating loses the information about constants, so we always recover it by substituting a convenient value of x, usually $x = 0$, into the original equation.

Solution

The derivative of an integral

Write $F(x) = \int_a^x \mathrm{f}(t)\,\mathrm{d}t$, the area under $y = \mathrm{f}(t)$ between $t = a$ and $t = x$. Increasing x by a small amount h adds a thin strip of width h:



The strip has area $F(x+h) - F(x)$. If f is continuous, then on $[x, x+h]$ the curve stays within a small amount $\varepsilon(h)$ of $\mathrm{f}(x)$, where $\varepsilon(h) \to 0$ as $h \to 0$. So the strip lies between rectangles of height $\mathrm{f}(x) \pm \varepsilon(h)$:

$$(\mathrm{f}(x) - \varepsilon(h))h \leq F(x+h) - F(x) \leq (\mathrm{f}(x) + \varepsilon(h))h.$$

Dividing by $h > 0$ (the case $h < 0$ is the same with the strip on the left) and letting $h \to 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x \mathrm{f}(t)\,\mathrm{d}t = \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \mathrm{f}(x).$$

Notice that a doesn't appear: changing the lower limit only adds a constant to F.

The equation $\mathrm{f}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t + 1$

The right-hand side is differentiable by the result above, so f is too, and differentiating both sides gives

$$\mathrm{f}'(x) = \mathrm{f}(x).$$

Putting $x = 0$ in the original equation, the integral is over an interval of zero length, so $\mathrm{f}(0) = 1$.

Ask yourself How do we know $\mathrm{f}' = \mathrm{f}$ has no solutions other than multiples of e^x?

Consider $g(x) = \mathrm{e}^{-x}\mathrm{f}(x)$. By the product rule,

$$g'(x) = \mathrm{e}^{-x}(\mathrm{f}'(x) - \mathrm{f}(x)) = 0,$$

so g is constant, and $g(0) = \mathrm{f}(0) = 1$. Hence $\mathrm{e}^{-x}\mathrm{f}(x) = 1$, that is, $\mathrm{f}(x) = \mathrm{e}^x$, as required. (Conversely, $\int_0^x \mathrm{e}^t\,\mathrm{d}t + 1 = \mathrm{e}^x - 1 + 1 = \mathrm{e}^x$, so it really is a solution.)

The equation $\mathrm{f}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$

Exactly as before, $\mathrm{f}' = \mathrm{f}$, so $\mathrm{e}^{-x}\mathrm{f}(x)$ is constant. But now $x = 0$ gives $\mathrm{f}(0) = 0$, so the constant is 0, and the only solution is

$$\mathrm{f}(x) = 0 \quad \text{for all } x.$$

The "+1" in the previous equation was what fixed the size of the exponential; without it only the zero function survives.

The final part

Differentiate both sides with respect to x. On the right, x is now the *lower* limit, and

$$\int_x^1 t^2\mathrm{f}(t)\,\mathrm{d}t = -\int_1^x t^2\mathrm{f}(t)\,\mathrm{d}t, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\int_x^1 t^2\mathrm{f}(t)\,\mathrm{d}t = -x^2\mathrm{f}(x).$$

Hence

$$\mathrm{f}(x) = -x^2\mathrm{f}(x) + 1 - x^4,$$
$$(1+x^2)\,\mathrm{f}(x) = 1 - x^4 = (1-x^2)(1+x^2).$$

Since $1 + x^2 > 0$ we may divide by it, giving

$$\mathrm{f}(x) = 1 - x^2.$$

To find C, put $x = 0$ in the given equation. The left-hand side and the terms $x - \frac{x^5}{5}$ vanish, leaving

$$0 = \int_0^1 t^2(1-t^2)\,\mathrm{d}t + C = \left[\tfrac{t^3}{3} - \tfrac{t^5}{5}\right]_0^1 + C = \tfrac{2}{15} + C,$$

so $C = -\frac{2}{15}$, as required.

Ask yourself Differentiating only tells us the two sides have the same derivative. Does $\mathrm{f}(x) = 1 - x^2$ with $C = -\frac{2}{15}$ really satisfy the equation for every x?

Yes. With this f the two sides have equal derivatives for all x, so they differ by a constant; and we chose C precisely so that they agree at $x = 0$. So they agree everywhere. Conversely, any solution must satisfy the differentiated equation, so $1 - x^2$ is the only possible f.

Checks

- At $x = 1$: the left side is $\int_0^1 (1-t^2)\,\mathrm{d}t = \frac{2}{3}$, and the right side is $0 + 1 - \frac{1}{5} - \frac{2}{15} = \frac{10}{15} = \frac{2}{3}$. They match.
- Numerically, Simpson's rule gives $\int_0^1 t^2(1-t^2)\,\mathrm{d}t \approx 0.133333 = \frac{2}{15}$, and both sides of the final equation agree to 10^{-10} at $x = -1.3$, 0.25, 0.5, 0.9, 2.

Common slips

- Forgetting the minus sign when the variable is the *lower* limit: $\frac{\mathrm{d}}{\mathrm{d}x}\int_x^1 g(t)\,\mathrm{d}t = -g(x)$.
- Solving $\mathrm{f}' = \mathrm{f}$ and stopping at Ae^x: the original integral equation, at $x = 0$, is what fixes A.
- Writing $\mathrm{f}(x) = \int_0^x \mathrm{f}(x)\,\mathrm{d}x$ style expressions that mix the dummy variable t with the limit x.

Beyond STEP

Equations in which the unknown function appears inside an integral are called *integral equations*; the ones here, with a variable limit, are Volterra equations, and they are equivalent to an initial-value problem for a differential equation. In a first analysis course you prove the Fundamental Theorem of Calculus carefully from the definition of the Riemann integral, and the strip argument above becomes an ε–δ proof. Abbott's *Understanding Analysis* does this very readably.

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